

Pair Of Lines

EE25BTECH11008 - Anirudh M Abhilash

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Problem Statement

If one of the lines of

$$my^2 + (1 - m^2)xy - mx^2 = 0$$

is a bisector of the angle between the lines

$$xy = 0,$$

then find the value of m .

Solution

The given homogeneous equation can be expressed as

$$(x \ y) \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0. \quad (1)$$

Let

$$A = \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix}.$$

Each line represented by this homogeneous equation passes through the origin, and its direction vector $\mathbf{v}$ satisfies

$$\mathbf{v}^T A \mathbf{v} = 0. \quad (2)$$

Solution (cont..)

For the bisectors of the coordinate axes ($xy = 0$), the directions are along $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Substituting $\mathbf{v}_1$,

$$\mathbf{v}_1^T A \mathbf{v}_1 = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (3)$$

$$= -m + (1 - m^2) + m = 1 - m^2. \quad (4)$$

Similarly, substituting $\mathbf{v}_2$,

$$\mathbf{v}_2^T A \mathbf{v}_2 = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} -m & \frac{1-m^2}{2} \\ \frac{1-m^2}{2} & m \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (5)$$

$$= -m - (1 - m^2) + m = m^2 - 1. \quad (6)$$

Solution (cont..)

For either $\mathbf{v}_1$ or $\mathbf{v}_2$ to represent a line of the pair,

$$1 - m^2 = 0. \quad (7)$$

Hence,

$$m = \pm 1. \quad (8)$$

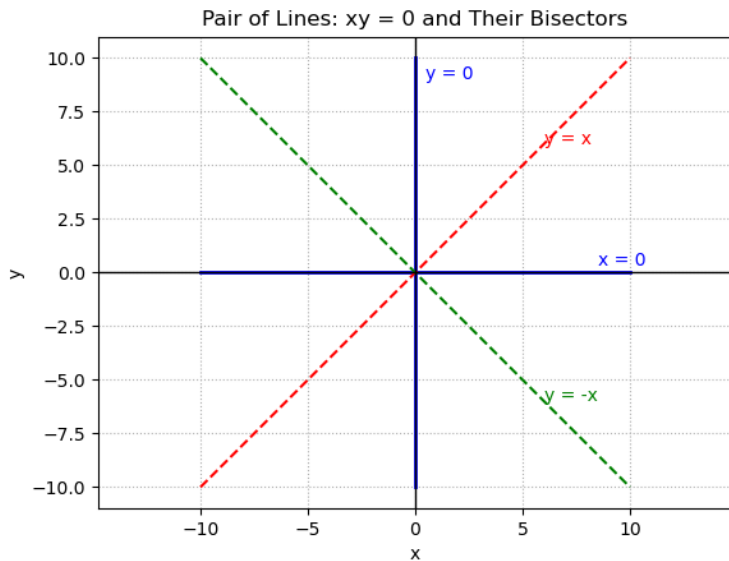
$$\boxed{m = \pm 1}$$

Python Code(Plot)

https:

`//github.com/Anirudh-EE25BTECH11008/ee1030-2025/tree/
main/EE25BTECH11008/MATGEO/4.13.13/Codes/plt.py`

Plot



C Code (Computations)

https:

`//github.com/Anirudh-EE25BTECH11008/ee1030-2025/tree/
main/EE25BTECH11008/MATGEO/4.13.13/Codes/verify.c`

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